

CRISTOFF STAN

Senior Software Engineer

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PROFESSIONAL SUMMARY

Deep Learning Scientist with over 9 years of experience in software engineering, specializing in biologically grounded AI systems for early drug discovery. Expert in designing and deploying GNNs, VAEs, and Transformers for target identification, MOA prediction, and patient stratification. Strong background in systems biology, multi-omics data integration (including scRNA-seq), and biological graph modeling. Passionate about advancing therapeutic pipelines through AI-driven innovation.

TECHNICAL SKILLS

- **Deep Learning & ML:** PyTorch, TensorFlow, JAX, scikit-learn, Hugging Face Transformers, DeepChem
- **Bioinformatics:** Scanpy, AnnData, Seurat (via R/Python interface), BioPython, scVI, OmicLearn
- **Models:** GNNs (GraphSAGE, GAT), VAEs, Transformers (BERT for biology), Autoencoders, UMAP
- **Data:** Multi-omics data (scRNA-seq, RNA-seq, ATAC-seq, proteomics), NGS pipelines
- **Other:** Docker, Git, Jupyter, NumPy, Pandas, NetworkX, FastAPI, REST APIs
- **Cloud & Infra:** Google Cloud Platform, AWS, Kubernetes (basic), Weights & Biases, MLFlow

WORK EXPERIENCE

Senior Software Engineer

Sensidev

September 2021 – December 2024

Timișoara, Romania (Remote)

- Designed and implemented VAEs and graph neural networks for molecular representation learning and bioactivity prediction across oncology-focused datasets.
- Developed pipeline to preprocess single-cell RNA-seq data using Scanpy, performing dimensionality reduction, clustering, and gene expression imputation.
- Integrated multi-omics data (RNA-seq, ATAC-seq, proteomics) for target prioritization using unsupervised deep learning architectures.
- Led design of biologically constrained Transformers for annotation of cell types and modeling perturbation responses.
- Collaborated with computational biologists to refine network biology models, improving pathway mapping and MOA prediction.
- Built a robust pipeline for patient stratification leveraging scVI, enabling precise subgrouping in immunotherapy datasets.

- Mentored junior engineers in applying machine learning best practices to biological data and reproducible research workflows.
- Delivered novel model architectures tailored for biological graph networks, contributing to internal publications and preprints.

Software Engineer

Syscom Digital

October 2019 – August 2021

Bucharest, Romania (Remote)

- Participated in development of deep learning infrastructure supporting omics data pipelines for clinical AI research projects.
- Contributed to implementation of ensemble GNN-based pipelines for target validation against real-world expression datasets.
- Automated data annotation and QC pipelines for single-cell transcriptomics using integrated Python-R workflows.
- Evaluated model robustness on diverse biological graphs involving protein interactions and gene regulatory networks.
- Assisted in cross-validation and benchmarking of drug response prediction models using VAEs and Bayesian networks.
- Streamlined ML pipeline deployment in containerized environments using Docker and GitLab CI/CD.
- Conducted literature review and model replication for academic benchmarks on multi-omics disease classification.
- Collaborated with biostatisticians to integrate classical survival models with learned deep representations.

Software Developer

Soft Build

June 2016 – July 2019

Timișoara, Romania (Remote)

- Engineered early generative models to simulate biological pathway activation patterns under perturbation.
- Created modular toolsets for omics data ingestion and exploratory analysis, including normalization, batch correction, and visualization.
- Developed RESTful APIs for scalable integration of biological data processing and ML workflows.
- Implemented real-time dashboards using Python backends for monitoring training convergence and biomarker trends.
- Applied probabilistic graphical models to infer latent biological variables and enhance pathway discovery.
- Worked with ontologies and biological databases (e.g., KEGG, Reactome, STRING) to augment graph-based learning.
- Optimized model training for sparse biological graphs using specialized data loaders and graph sampling methods.
- Participated in hackathons focused on AI for health, winning 2nd prize for gene-disease association prediction using GNNs.

Software Developer
Decima Digital
November 2015 – April 2016
Tallinn, Estonia

- Supported implementation of full-stack interfaces for biological research tools, enabling visualization of large-scale cell-type maps.
- Assisted in building data parsers and annotation tools for next-generation sequencing datasets.
- Collaborated with senior ML scientists to validate deep learning approaches for transcription factor activity inference.
- Refactored Python codebases for omics feature engineering, increasing pipeline efficiency by 30%.
- Designed test suites for model explainability metrics including SHAP and integrated gradients on biological inputs.
- Applied PCA and t-SNE visualizations to biological embeddings for expert review and interpretation.
- Built modular scripts to automate graph construction from protein-protein interaction datasets.
- Explored integration of medical imaging data with genomic markers in preliminary research trials.

Software Developer Intern
Borna Technology
September 2014 – August 2015
Tallinn, Estonia (On-Site)

- Created Python scripts to preprocess public bioinformatics datasets (e.g., GEO, TCGA) for downstream modeling.
- Learned and applied fundamentals of neural network architectures on toy biology datasets for drug activity prediction.
- Supported design of internal tools for feature selection across high-dimensional omics data.
- Developed web interface prototypes to visualize clustering results of gene expression profiles.
- Conducted exploratory data analysis and implemented basic classifiers (RF, SVMs) for disease prediction.
- Participated in weekly lab meetings discussing computational biology concepts, tools, and strategies.
- Assisted in automating literature extraction using NLP for gene-drug associations.
- Implemented multi-view clustering pipelines integrating proteomics and transcriptomics features.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,
2010 – 2014

CERTIFICATIONS

- ❖ Deep Learning Specialization – Andrew Ng (Coursera)
- ❖ Graph Neural Networks for Researchers – Stanford CS224W
- ❖ AI for Medicine Specialization – Coursera (deeplearning.ai)
- ❖ Bioinformatics Specialization – UC San Diego (Coursera)
- ❖ Single-cell RNA-seq Analysis – HarvardX (edX)

PROJECTS

- ✓ **GraphDrugAI** – Deep Learning Framework for Target Discovery
Designed a suite of GNN-based models for identifying novel drug targets from multi-omics data. Integrated Reactome pathway graphs with transcriptomic embeddings to prioritize targets in cancer datasets.
- ✓ **OmicsStratifier** – Patient Stratification via scVI and GNNs
Built pipeline using scVI and biological GAT models to stratify patients based on single-cell RNA-seq data. Applied to immunotherapy response cohorts to identify latent responder groups.
- ✓ **TransMOA** – Transformer-based MOA Prediction
Developed hybrid Transformer-VAE models for predicting mechanism of action across multi-modal biological datasets. Achieved significant improvements over baseline models on benchmark datasets (LINCS, CMAP).
- ✓ **BioGraphX** – Network Biology Toolkit
Open-source toolset combining systems biology models and deep learning for drug repurposing. Includes modules for gene network extraction, biological graph encoding, and causal inference modeling.